

Prediction of Human Oral Bioavailability Using Machine and Deep Learning, and Model Interpretation with SHAP Values

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In brief. A QSAR workflow predicts whether a small molecule has high or low human oral bioavailability (HOB) from its structure, and explains each prediction. Tree ensembles reached ROC-AUC 0.820–0.895 on two external test sets at the 50% threshold. 2D descriptors carry most of the predictive signal, while 3D descriptors contribute slightly more to the model's explanations than their share of the feature set.

Background and aim

Oral bioavailability is the fraction of an oral dose that reaches the systemic circulation unchanged. It is a critical pharmacokinetic property, and measuring it experimentally is costly and slow. The thesis builds a predictive framework for HOB with an emphasis on interpretability, and addresses three questions:

- Which class of molecular descriptors, 2D or 3D, is more predictive of HOB, and what does combining them add?
- Which machine learning or deep learning models perform best, judged on independent test sets?
- What do SHAP values reveal about the relationship between molecular structure and bioavailability?

Data

The public HobPre dataset (Wei et al., 2022). HOB is treated as a binary label under two definitions: a 50% cutoff (nearly balanced) and a 20% cutoff (imbalanced).

Subset	Molecules	High / Low at 50%	High / Low at 20%
Training set	1,142	621 / 521	859 / 283
Test set 1	290 (287 at 20%)	169 / 121	214 / 73
Test set 2	140 (132 at 20%)	89 / 51	127 / 5

Methods

- **Descriptors.** 3D conformers generated with RDKit (ETKDGv3, MMFF). Mordred computes 1,826 descriptors (1,613 2D, 213 3D); 1,672 remain after missing-value handling.
- **Feature selection.** Two methods compared: the intersection of mutual-information and CatBoost rankings (243 descriptors at 50%, 232 at 20%) and Elastic Net (318 and 346).
- **Models.** Seven classifiers screened and combined in soft-average ensembles, scored by ROC-AUC with stratified 5-fold cross-validation.
- **Evaluation.** The decision threshold is set on training predictions (Youden's J). The two test sets are used once, after all choices are fixed.
- **Neural network.** A multilayer perceptron on all 1,672 descriptors, tuned over 27 configurations, averaged over three seeds.
- **Interpretation.** SHAP values (TreeExplainer for the tree models, expected gradients for the network), including the share attributed to 2D and 3D descriptors.

Results

Cutoff	Final model	Test set	n	ROC-AUC	MCC	Bal. acc.	Sens.	Spec.	MLP ROC-AUC
50%	ExtraTrees + CatBoost	Test set 1	290	0.820	0.478	0.742	0.716	0.769	0.794
50%	ExtraTrees + CatBoost	Test set 2	140	0.895	0.629	0.822	0.820	0.824	0.879
20%	ExtraTrees	Test set 1	287	0.830	0.456	0.755	0.743	0.767	0.779
20%	ExtraTrees	Test set 2*	132	0.981	0.366	0.902	0.803	1.000	0.943

* Only five Low molecules; class-specific metrics for this case rest on five observations. MLP: multilayer perceptron evaluated on the same molecules.

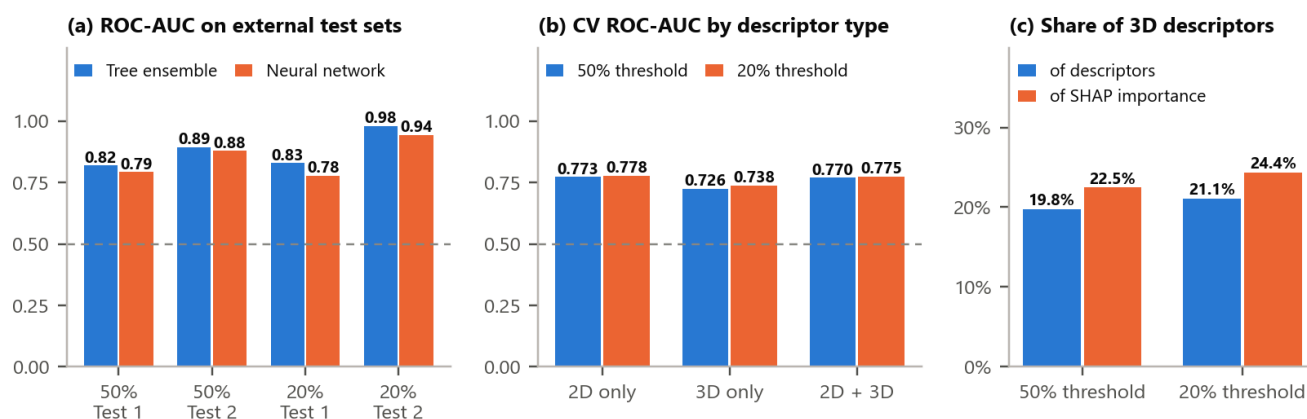


Figure. (a) Final tree models and the neural network on the external test sets; dashed line = chance. (b) Final model trained on the 2D part, the 3D part, or the whole selected set. (c) 3D descriptors as a share of the selected set and of the total mean |SHAP| (training set).

Key findings

- **Tree ensembles performed best.** ExtraTrees + CatBoost (50%) and ExtraTrees (20%) were selected by cross-validation (ROC-AUC about 0.77) and scored higher than the neural network in all four external evaluations.
- **2D descriptors carry most of the signal.** 2D alone matched the full set (0.773 vs 0.770 at 50%); 3D alone was lower (0.726). The ordering held at equal set sizes.
- **3D descriptors matter for explanation.** They are 19.8% and 21.1% of the selected sets but 22.5% and 24.4% of total SHAP importance.
- **The two feature selection methods are equivalent in accuracy.** They differ by about 0.01 ROC-AUC or less; MI $\cap$ CatBoost needs fewer descriptors.
- **Explanations agree with known chemistry.** SHAP highlights polar surface area, hydrogen-bonding capacity and molecular size, consistent with established structure–activity relationships.

Limitations

- All data come from one curated source of 1,142 training molecules; performance on other chemical spaces is untested.
- Test-set scores are higher than cross-validated scores, so the test sets may be easier than the training distribution. Test set 2 at the 20% cutoff has only five Low molecules.
- SHAP describes what the model relies on. It shows association with HOB, not causal mechanisms.

Conclusion

Interpretable tree ensembles on Mordred descriptors predict HOB class at a level comparable to published results on the same data, and their explanations match established medicinal-chemistry knowledge. This makes the predictions checkable and usable as guidance when prioritising or modifying candidate molecules.

Tools: Python, RDKit, Mordred, scikit-learn, CatBoost, SHAP, Keras/TensorFlow. **Data:** Wei, M. et al. HobPre: accurate prediction of human oral bioavailability for small molecules. *Journal of Cheminformatics* 14 (2022).